Git Installation Guide

Complete Step-by-Step Setup for All Operating Systems

Data Science Club · USTHB University

Web Dev Section — Core Team — *M.Souhil*

Choose Your Operating System

Click the button that matches your system — the guide jumps directly to your section.

Linux
Click to jump

macOS
Click to jump

Windows
Click to jump

After installing Git, go to Section 6 to create your GitHub account.

Version: 1.0	**Reading time:** ≈30 min
Audience: Beginners	**Level:** Introductory
Year: 2025–2026	**Contact:** mokeddemsouhil968@gmail.com

This document is intended for club members.
Covers Git installation on Linux, macOS, and Windows, plus GitHub account creation.

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What is Git and Why Do We Need It?

Git is a **version control system**. Think of it as a super-powered "undo" button for your code, combined with a tool that lets **multiple people work on the same project** without overwriting each other's changes.

Why Git Matters for You

As a member of the DS Club web development section, every project you contribute to will use Git. Here is why it is indispensable:

- ✔ **Track every change** — go back to any previous version at any time.
- ✔ **Collaborate** with teammates without conflicts or overwritten files.
- ✔ **Upload** your code to GitHub so others can see, review, and contribute.
- ✔ **Industry standard** — used by virtually every tech company worldwide.

Git is the tool installed on your computer.
GitHub is a website (github.com) that stores your Git repositories online.
You need **both**: Git first (this guide), then a GitHub account (Section 6).

Prerequisites — Check Before You Install

- ☑ **A working computer** running Linux, macOS, or Windows.
- ☑ **Internet connection** — required to download the installer.
- ☑ **Administrator/sudo privileges**.
  Windows: right-click → *Run as administrator*.
  Linux/macOS: you will use `sudo` (explained in each section).
- ☑ **About 500 MB of free disk space**.
- ! **Close antivirus temporarily** on Windows if it blocks the installer.

Installing Git on Linux

Linux — Git Installation Guide

Linux is the most straightforward system for installing Git. We cover **APT** (Ubuntu, Debian, Linux Mint) and **DNF/YUM** (Fedora, RHEL, CentOS).

STEP 1 · Open the Terminal

- **Ubuntu/Linux Mint:** Press `Ctrl + Alt + T`
- **Most desktops:** Right-click on the desktop → *Open Terminal*

Terminal - Ubuntu 22.04

```
user@ubuntu:~$
```

STEP 2 · Update the Package List

Ubuntu / Debian / Linux Mint -- APT

```
sudo apt update
```

Fedora -- DNF

```
sudo dnf check-update
```

CentOS / RHEL -- YUM

```
sudo yum check-update
```

`sudo` gives you temporary administrator powers. **Nothing will appear on screen while you type your password** — this is normal on Linux.

Terminal - after running sudo apt update

```
user@ubuntu:~$ sudo apt update
sudo
password for user:
Hit:1 https://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 https://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:3 https://archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Fetched 2,453 kB in 1s (2,453 kB/s)
Reading package lists... Done
```

STEP 3 · Install Git

>_ Ubuntu / Debian / Linux Mint -- APT

```
sudo apt install git -y
```

>_ Fedora -- DNF

```
sudo dnf install git -y
```

>_ CentOS / RHEL -- YUM

```
sudo yum install git -y
```

Terminal - sudo apt install git -y

```
user@ubuntu:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
The following NEW packages will be installed:
git  git-man  liberror-perl
0 upgraded, 3 newly installed, 0 to remove.
Get:1 .../git_1:2.43.0-1ubuntu1_amd64.deb [3,164 kB]
Get:2 .../git-man_1:2.43.0-1ubuntu1_all.deb [946 kB]
Fetched 4,110 kB in 2s (2,055 kB/s)
Unpacking git (1:2.43.0-1ubuntu1) ...
Setting up git (1:2.43.0-1ubuntu1) ... Done
```

STEP 4 · Verify the Installation

>_ Check Git version

```
git -version
```

Terminal - git -version

```
user@ubuntu:~$ git -version
git version 2.43.0
```

git version 2.43.0

Any version above **2.30** is great.

Try running the install command again, or check your internet connection.

STEP 5 · Configure Git with Your Identity

Set your name and email

```
git config -global user.name "Your Full Name" git config -global
user.email "your.email@example.com"
```

Verify the configuration was saved

```
git config -list
```

Terminal - git config -list

```
user@ubuntu:~$ git config -list
user.name=Your Full Name
user.email=your.email@example.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
```

Quick Test — Make Sure Git Actually Works

Create a test repository

```
mkdir git-test-club cd git-test-club git init
```

Create a file and commit

```
echo "Hello DS Club!" > hello.txt git add hello.txt git commit -m "My first
Git commit"
```

Terminal - full test run

```
user@ubuntu:~$ mkdir git-test-club
user@ubuntu:~$ cd git-test-club
user@ubuntu:~/git-test-club$ git init
Initialized empty Git repository in /home/user/git-test-club/.git/
user@ubuntu:~/git-test-club$ echo "Hello DS Club!" > hello.txt
user@ubuntu:~/git-test-club$ git add hello.txt
user@ubuntu:~/git-test-club$ git commit -m "My first Git commit"
[main (root-commit) a3f9c12] My first Git commit
1 file changed, 1 insertion(+)
create mode 100644 hello.txt
```

You just created your first Git repository and made your first commit. Jump to Section 6 to create your GitHub account.

→ **Done with Linux?** Jump to: **Section 6 — Create Your GitHub Account**

Installing Git on macOS

macOS — Git Installation Guide

On macOS you have **two options**. We recommend **Option A (Homebrew)**. Option B uses Apple's built-in developer tools.

Option A — Install via Homebrew (Recommended)

STEP 1 · Open the Terminal

- Press `Command + Space`, type **Terminal**, press **Enter**
- Or: *Finder* → *Applications* → *Utilities* → *Terminal*

```
Terminal - macOS (zsh)
username@MacBook-Pro ~ %
```

STEP 2 · Install Homebrew (Package Manager)

Copy and paste this entire command into your terminal (one single line):

Install Homebrew -- paste this EXACTLY

```
/bin/bash -c "(curl−fsSLhttps://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)
```

Terminal - Homebrew installation in progress

```
username@MacBook-Pro ~ % ...
==> Checking for sudo access (may request password)...
Password:
==> This script will install Homebrew to /opt/homebrew
==> Downloading Command Line Tools for Xcode...
==> Installing Command Line Tools for Xcode...
==> Downloading and installing Homebrew...
==> Fetching homebrew/homebrew-core...
==> Homebrew is ready.  Installation successful!
==> Run `brew help` to get started
```

https://brew.sh

After Homebrew installs, run this extra command:

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zshrc
eval "$(/opt/homebrew/bin/brew shellenv)"
```

Not sure which chip you have? Click Apple menu → *About This Mac.*

STEP 3 · Install Git via Homebrew

Install Git with Homebrew

```
brew install git
```

Terminal - brew install git

```
username@MacBook-Pro ~ % brew install git
==> Fetching dependencies:  gettext, pcre2
==> Fetching git
==> Downloading https://formulae.brew.sh/api/formula/git
################################# 100.0%
==> Installing git
==> Pouring git-2.43.0.arm64_ventura.bottle.tar.gz
/opt/homebrew/Cellar/git/2.43.0:  1,593 files, 49.6MB
```

STEP 4 · Verify the Installation

Check Git version

```
git -version
```

Terminal - git -version on macOS

```
username@MacBook-Pro ~ % git -version
git version 2.43.0
```

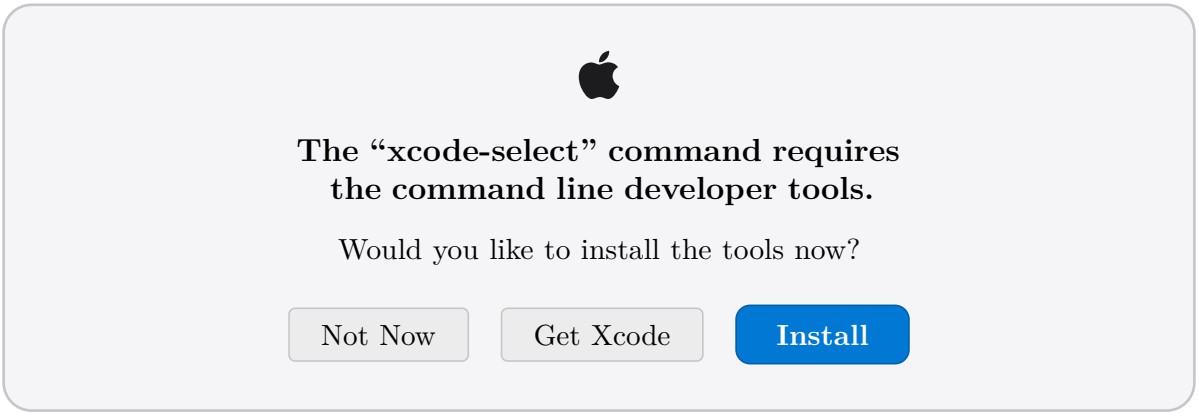
Option B — Install via Xcode Command Line Tools

STEP 5 · Run this command in Terminal

Trigger Xcode Command Line Tools installer

```
xcode-select -install
```

A dialog box will appear on your screen. Click **Install**:



*Click **Install** — NOT "Get Xcode". The download is about 200 MB.*

STEP 6 · Verify after Xcode Tools install

Check Git version

```
git -version
```

```
git version 2.39.3 (Apple Git-145)
```

The "Apple Git" note is fine — it means Apple's bundled version.

Configure Git — Required for Both Options

STEP 1 · Set Your Identity

Set name and email

```
git config -global user.name "Your Full Name" git config -global
user.email "your.email@example.com"
```

Verify configuration

```
git config -list
```

Quick Test — macOS

Run this test -- line by line

```
mkdir git-test-club cd git-test-club git init echo "Hello DS Club!" >
hello.txt git add hello.txt git commit -m "My first Git commit"
```

Terminal - full test run on macOS

```
username@MacBook-Pro ~ % mkdir git-test-club
username@MacBook-Pro ~ % cd git-test-club
username@MacBook-Pro git-test-club % git init
Initialized empty Git repository in    /Users/username/git-test-club/.git/
username@MacBook-Pro git-test-club % git commit -m "My first Git commit"
[main (root-commit) b7d1e43] My first Git commit
1 file changed, 1 insertion(+)
```

You are ready to create your GitHub account in Section 6.

→ **Done with macOS?** Jump to: **Section 6 — Create Your GitHub Account**

Installing Git on Windows

Windows — Git Installation Guide

On Windows we install **Git for Windows**, which includes:

- **Git** — the core version control tool
- **Git Bash** — a Unix-style terminal for Git commands
- **Git GUI** — a graphical interface (optional)

STEP 1 · Download from the Official Git Website

Open your browser and go to:

https://git-scm.com/download/win

https://git-scm.com/download/win

Git for Windows

Downloading Git 2.43.0 for Windows...

Git-2.43.0-64-bit.exe (56 MB)

The download should start automatically.
Click here if it does not start.

64-bit Git for Windows Setup | 32-bit Git for Windows Setup | Portable

Most modern computers need **64-bit**.
To check: Right-click *This PC* → *Properties* → look for *System type*.

STEP 2 · Launch the Installer — User Account Control

Find the `.exe` file in Downloads and **double-click** it.

User Account Control

Do you want to allow this app to
make changes to your device?

App: Git for Windows 2.43.0
Publisher: The Git Development Community

Yes No

*Click **Yes** to allow the installer to run.*

If Windows shows "Windows protected your PC", click **More info** then **Run anyway**. This is normal for new software.

STEP 3 · License Agreement Screen

Git Setup — License

GNU GENERAL PUBLIC LICENSE
Version 2, June 1991

Copyright (C) 1989, 1991 Free Software Foundation, Inc.
51 Franklin Street, Fifth Floor, Boston, MA 02110-1301 USA

Everyone is permitted to copy and distribute verbatim copies of this
license document, but changing it is not allowed...

Back Next Cancel

*Just click **Next**. You do not need to read all of this.*

STEP 4 · Installation Location

Git Setup — Select Destination Location

Please specify the folder where Git will be installed.

```
C:\Program Files\Git
```

At least 320 MB of free disk space is required.

Back | **Next** | Cancel

*Leave as default and click **Next**.*

STEP 5 · Select Components

Git Setup — Select Components

Which components should be installed?

- [x] **Windows Explorer integration**
  - [x] Git Bash Here
  - [x] Git GUI Here
- [x] **Associate .git* files with default text editor**
- [x] Add a Git Bash Profile to Windows Terminal
- [ ] Check daily for Git for Windows updates

Back | **Next** | Cancel

*Keep the defaults and click **Next**.*

STEP 6 · Choosing the Default Editor — IMPORTANT

Git Setup — Default Editor

Which editor would you like Git to use?

Current selection:

▾ Use Notepad as Git's default editor

Other available editors: Vim, Nano, VS Code, Notepad++...

Back | **Next** | Cancel

*Select **Notepad** from the dropdown, then click **Next**.*

The default is *Vim*, which is powerful but confusing for beginners. Notepad is much friendlier.

STEP 7 · Initial Branch Name

Git Setup — Initial Branch

What would you like to call the initial branch in a new repository?

○ Let Git decide (currently: master)

◉ **Override the default branch name**

`main`

Back | **Next** | Cancel

*Select Override and type **main**. Click **Next**.*

STEP 8 · Adjusting PATH Environment — IMPORTANT

Git Setup — Adjusting your PATH environment

How should Git be used from the command line?

- Use Git from Git Bash only
- **Git from the command line and also from 3rd-party software**
  Recommended: Git will be added to your PATH.
- Use Git and optional Unix tools from CMD

Back | **Next** | Cancel

*Select the **middle option** (recommended) and click **Next**.*

STEP 9 · SSH, HTTPS, Line Endings, Terminal — Next through each

For the next four screens, keep the defaults shown and click **Next** each time:

- **SSH executable:** Use bundled OpenSSH → **Next**
- **HTTPS backend:** Use the OpenSSL library → **Next**
- **Line endings:** Checkout Windows-style, commit Unix-style → **Next**
- **Terminal emulator:** Use MinTTY → **Next**

Git Setup — Line Ending Conversions

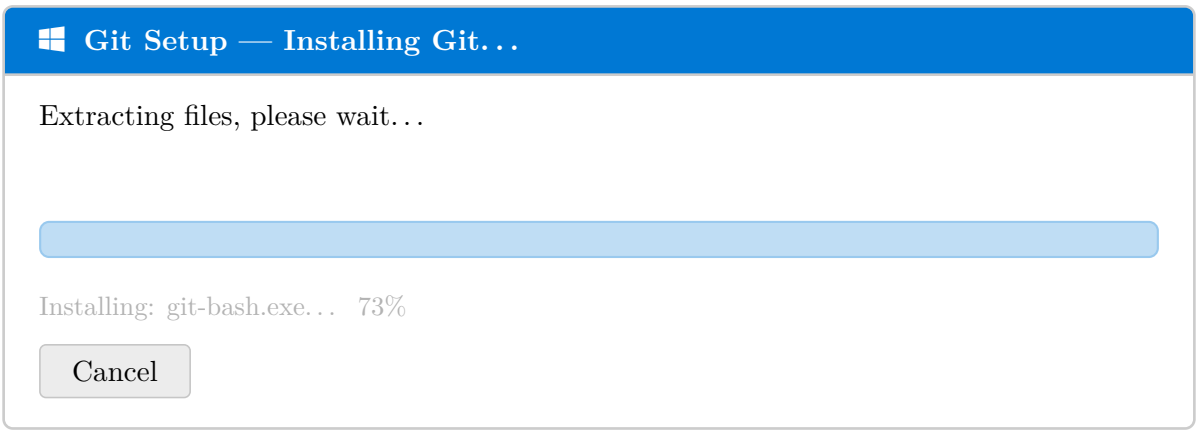
How should Git treat line endings in text files?

- **Checkout Windows-style, commit Unix-style line endings**
  Recommended for Windows users.
- Checkout as-is, commit Unix-style line endings
- Checkout as-is, commit as-is

Back | **Next** | Cancel

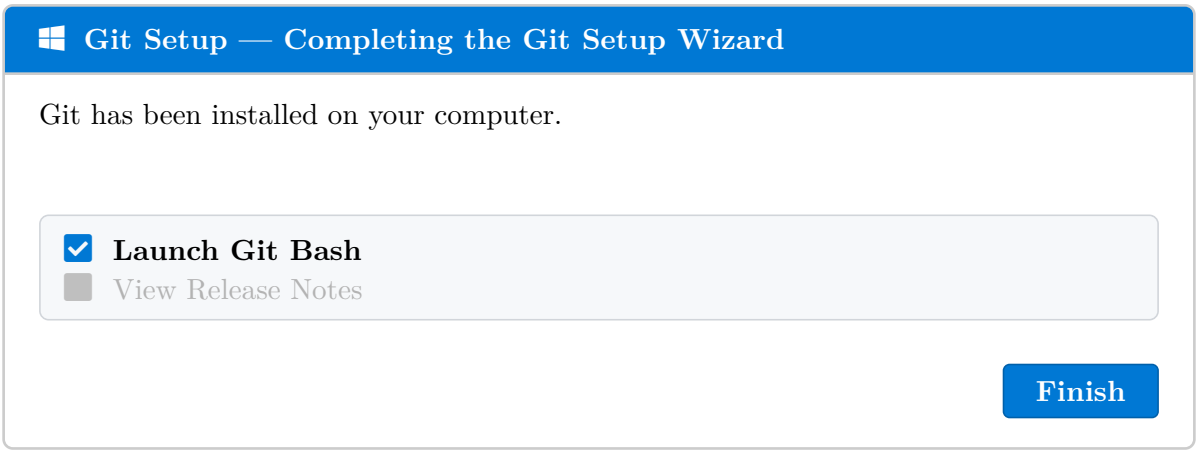
STEP 10 · git pull, Credential Helper, Extra Options — Next

- **git pull behavior:** Default (fast-forward or merge) → **Next**
- **Credential helper:** Git Credential Manager → **Next**
- **Extra options:** Enable file system caching (checked) → **Next**
- **Experimental:** Leave all unchecked → **Install**



The progress bar fills up. Wait 1–3 minutes.

STEP 11 · Installation Complete



*Check **Launch Git Bash** and click **Finish**.*

Verify the Installation on Windows

A Git Bash window will open. Type:

```
Verify Git installation
git -version
```

```
Git Bash - Windows
MINGW64 ~
$ git -version
git version 2.43.0.windows.1
```

```
git version 2.43.0.windows.1
```

Git is installed and ready!

Configure Git with Your Identity (Windows)

Set your name and email

```
git config -global user.name "Your Full Name" git config -global user.email
"your.email@example.com"
```

Verify settings

```
git config -list
```

```
user.name=Your Full Name
user.email=your.email@example.com
```

Quick Test — Windows

Run these commands **one by one** in Git Bash:

Create and test a Git repository

```
mkdir git-test-club cd git-test-club git init echo "Hello DS Club!" >
hello.txt git add hello.txt git commit -m "My first Git commit"
```

```
Git Bash - Windows test
MINGW64 ~
$ mkdir git-test-club
$ cd git-test-club
$ git init
Initialized empty Git repository in
C:/Users/YourName/git-test-club/.git/
$ git commit -m "My first Git commit"
[main (root-commit) d2c5a89] My first Git commit
1 file changed, 1 insertion(+)
```

Jump to Section 6 to create your GitHub account.

- **Search:** Start → type "Git Bash" → Enter
- **Right-click** any folder → *Git Bash Here*

→ **Done with Windows?** Jump to: **Section 6 — Create Your GitHub Account**

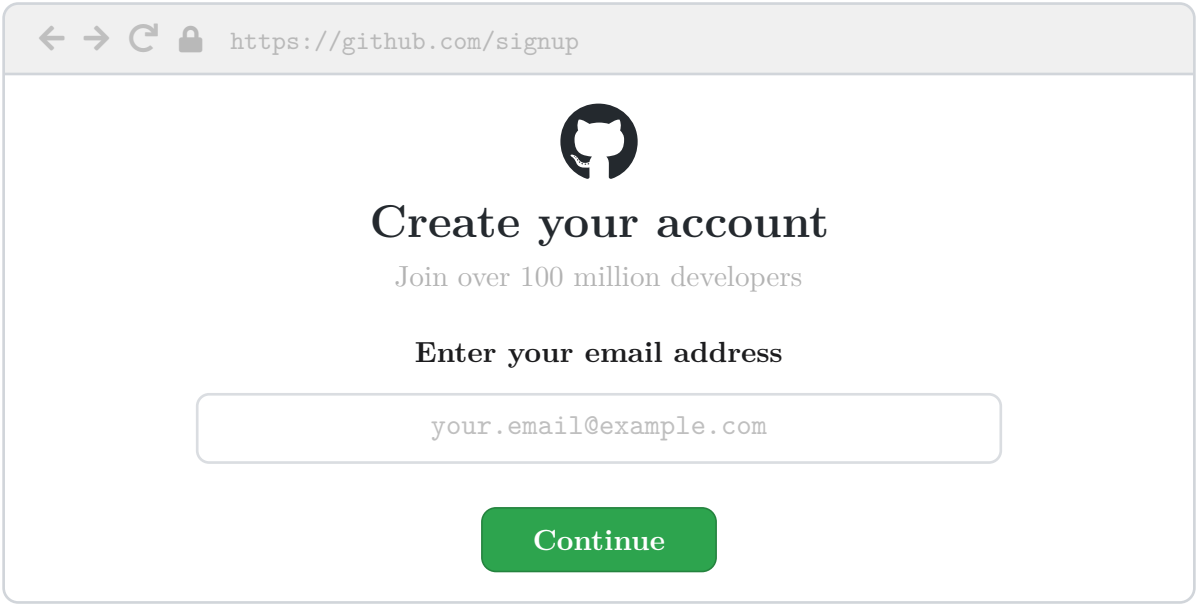
Creating Your GitHub Account

GitHub — Create Your Free Account

Now that Git is installed, you need a **GitHub account** to store your projects online and collaborate with club members. GitHub is completely **free** for individual use.

STEP 1 · Go to GitHub's Sign-Up Page

https://github.com/signup



*Enter your email address and click **Continue**.*

STEP 2 · Create a Password

Type a strong password (at least 15 characters, or 8+ with mixed types). Click **Continue**.

STEP 3 · Choose Your Username — IMPORTANT

Recommended format for club members:

- `firstname-lastname` — e.g., `souhil-merrouche`
- `firstnamelastname` — e.g., `souhilmerrouche`
- Keep it **professional** (no random numbers like `user99`)
- Use only **letters, numbers, and hyphens**

https://github.com/signup

Username

souhil-merrouche

✔ souhil-merrouche is available!

gamer99

✖ Username is already taken. Try another.

● *Available* ● *Taken. Click **Continue** when you find a free one.*

STEP 4 · Human Verification

GitHub shows a simple puzzle. Solve it to confirm you are not a robot.

https://github.com/signup

Verify your account

Solve the puzzle to continue

Try a new puzzle

*Complete the puzzle, then click **Continue**.*

STEP 5 · Enter the Verification Code

GitHub sends a **6-digit code** to your email. Check your inbox and enter it:

https://github.com/signup

Enter the code sent to your email

We sent a launch code to your.email@example.com

3 7

Resend code

Check your **spam folder**, wait 2–3 minutes, or click *Resend code*.

STEP 6 · Choose a Plan

https://github.com/signup

Choose your plan

Free
$0/month
- Unlimited public repositories
- Unlimited private repositories
- 2,000 Actions minutes/month

Continue for free

Pro
$4/month
- Everything in Free
- Advanced code review
- GitHub Copilot (paid)

Choose Pro

*Click **Continue for free**. The free plan is everything you need.*

Your profile is live at:

`https://github.com/YOUR-USERNAME`

Finding Your GitHub Username

How to Find and Share Your GitHub Username

The team needs your GitHub username to add you to club repositories.

Method 1 — From Your Profile Page

STEP 7 · Click Your Profile Picture (top-right corner)

https://github.com

souhil

Signed in as **@souhil-merrouche**

Your profile
Your repositories
Settings

*Click **Your profile** from the dropdown.*

STEP 8 · Find the Username on Your Profile

https://github.com/souhil-merrouche

Souhil Merrouche
@souhil-merrouche
Data Science Club — USTHB
Followers: 5 Following: 3

URL bar:

https://github.com/souhil-merrouche

Your username appears in the URL bar and below your name as `@username`.

Method 2 — From the Terminal

Check stored Git config

```
git config -global user.name
```

`user.name` in Git is your display name — it might not be the same as your GitHub username. Always verify on the website using Method 1.

Share Your Username with the Club

Your GitHub profile URL:

https://github.com/*your-username*

Share this URL with the team to be added to club projects.

Troubleshooting — Common Problems and Solutions

Cause: Git is not installed, or the terminal cannot find it.

Solution:

1. Try running the install command again.
2. On macOS: run `brew install git` again.
3. On Linux: run `sudo apt install git -reinstall`
4. Close and reopen your terminal, then try `git -version`.

Cause: Git not added to PATH, or you are using Command Prompt.

Solution:

1. Use **Git Bash** (not Command Prompt or PowerShell).
2. Reinstall Git and on the PATH screen choose *Git from the command line and also from 3rd-party software.*
3. Restart your computer after reinstalling.

Cause: Network issue, or Xcode Command Line Tools missing.

Solution:

1. First run `xcode-select -install` and complete it.
2. Then try the Homebrew install command again.
3. If your network is slow, it can take up to 30 minutes.

Cause: Typed password incorrectly.

Solution: When typing your password in the terminal, **nothing appears on screen** — not even dots. Type carefully and press **Enter**.

Cause: Name and email not configured.

Solution:

```
git config -global user.name "Your Full Name"
git config -global user.email "your.email@example.com"
```

Quick Reference Card

Most Important Git Commands

Command	What it does
`git init`	Creates a new Git repository in the current folder
`git status`	Shows the current state of your files
`git add .`	Stages all changed files to be committed
`git add filename`	Stages one specific file
`git commit -m "msg"`	Saves a snapshot with a description
`git log`	Shows the history of all commits
`git clone [url]`	Downloads a repository from GitHub
`git push`	Uploads your commits to GitHub
`git pull`	Downloads the latest changes from GitHub
`git -version`	Shows the installed Git version
`git config -list`	Shows all your Git settings

Useful Links and Resources

Resource	Link
Official Git website	https://git-scm.com
Git download (Windows)	https://git-scm.com/download/win
Git download (macOS)	https://git-scm.com/download/mac
Git download (Linux)	https://git-scm.com/download/linux
Homebrew (macOS)	https://brew.sh
GitHub sign up	https://github.com/signup
GitHub documentation	https://docs.github.com
Pro Git Book (free)	https://git-scm.com/book/en/v2
GitHub Student Pack	https://education.github.com/pack
Git cheat sheet	education.github.com/git-cheat-sheet-education.pdf

As a university student, you qualify for the **GitHub Student Developer Pack** — completely free! Use your university email (`@usthb.dz`) at education.github.com/pack to get GitHub Copilot, private repos, and many more tools.

Final Checklist — Did You Do Everything?

Complete this before your first club session

- Git is installed and `git -version` shows a version number.
- I set my **name**: `git config -global user.name "My Name"`
- I set my **email**: `git config -global user.email "me@mail.com"`
- I ran `git config -list` and saw my name and email.
- I completed the **quick test**: created a repo, added a file, and committed it.
- I created a **GitHub account** at github.com/signup.
- I know my GitHub **username** and shared it with the team.
- I bookmarked the GitHub Student Pack: education.github.com/pack

You are all set!

Welcome to the Data Science Club Web Dev team.
Your first real Git workflow is just one `git clone` away.

Questions? Reach out to the core team in the club channel.

Data Science Club · USTHB University · Web Development Section